

Mariana Figueiredo

Biological Engineer

 Mariana Figueiredo |  mdfigueiredop@gmail.com |  +351 914 665 196 |  Viseu, Portugal

SUMMARY

Results-driven Biological Engineer with +2 years of cumulative experience in industrial and laboratory settings through curricular internships. Focused on the practical application of bioengineering to solve real-world food technology and health problems. Seeking a challenging entry-level professional position to contribute to cutting-edge industry projects.

AREAS OF EXPERTISE

- | | | |
|------------------------|----------------------------|-------------------------------|
| ✓ Alternative Proteins | ✓ Bioactive Peptides | ✓ Design of Experiments (DoE) |
| ✓ Functional Foods | ✓ Interfacial Science | ✓ Enzymatic Bioprocessing |
| ✓ Data Analysis | ✓ Nutrient Bioavailability | ✓ Downstream Processing |

PROFESSIONAL EXPERIENCE

MSc Graduate Researcher

Feb 2026 - present

Laboratory for Process Engineering, Environment, Biotechnology and Energy

Porto, Portugal

- Engineering an immobilized-protease biocatalyst, optimized via **Design of Experiments (DoE)**, to valorize macroalgae side-streams into high-value bioactive peptides.

Research Intern

Sep 2025 - Jan 2026

Laboratory of Food Chemistry and Biochemistry KU Leuven

Leuven, Belgium

- Evaluated the **foaming properties** of **plant-based proteins** through standardized foaming assays.
- Characterized **interfacial properties** using tensiometry assays to correlate adsorption kinetics and interfacial rheology with foaming performance.
- Enhanced **sustainability profile** of sponge cakes by reducing egg reliance by 25% through plant-based protein substitution while maintaining batter and cake integrity.

Research Intern

Feb 2025 - Jun 2025

Laboratory for Process Engineering, Environment, Biotechnology and Energy

Porto, Portugal

- Extracted and enzymatically hydrolysed **macroalgae-based proteins** to yield bioactive peptides.
- Evaluated **anti-hypertensive** potential through ACE-inhibitory activity assays, achieving a IC_{50} of 52.4 $\mu\text{g/mL}$ and 56.6 $\mu\text{g/mL}$ for *Ulva* sp. and *Mastocarpus stellatus* peptides.
- Identified a total of 30 different potential anti-hypertensive peptides through an *in silico* screening and simulated their bioaccessability and bioavailability.

Engineer Intern

Feb 2024 - May 2024

AgroGrIN Tech

Porto, Portugal

- Contributed to the **optimization of production workflows**, identifying potential bottlenecks and proposing alternative unit operations.
- Conducted **equipment research** and feasibility studies for certain unit operations to assess their potential integration in the company's production system.

PROJECTS

Industrial Production of Alpha-Mannosidase

Fev 2025 - Jun 2025

Faculty of Engineering of University of Porto

- Developed a **patent-based industrial design** for alpha-mannosidase production and purification for pharmaceutical use.
- Managed the **project lifecycle** from initial market analysis and process description to process flow modeling, technical equipment sizing for upstream and downstream operations and economic analysis.

EDUCATION

Master’s Degree in Bioengineering

Sep 2024 - present

University of Porto

Bachelor’s Degree in Bioengineering

Sep 2021 - Jun 2024

University of Porto

ECTS Grade: B

VOLUNTEERING

Member of the Mentoring Department

Oct 2023 - Oct 2024

Já T’Explico

Responsible for planning and providing **weekly educational guidance** to children in need.

PROFESSIONAL DEVELOPMENT

Working with Clients

2024

British Council | Online

Foundations of Project Management

2024

Google (Coursera) | Online

From Basic to Advanced - Complete Microsoft Excel Course

2023

Udemy | Online

ADDITIONAL INFORMATION

Technical Proficiencies: LaTeX | JMP Software | MATLAB | Microsoft Excel

Languages: Portuguese | English